

Design, Control, and Evaluation of a Multijoint Lower Limb Soft Exosuit

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Abstract—Soft exosuits offer distinct advantages for human motion assistance; however, multijoint actuation strategies frequently encounter challenges associated with actuator redundancy. Conventional designs, typically employing independent actuator and Bowden cable configurations per joint, impose substantial mass penalties that undermine the inherent lightweight potential of soft exosuit technology. This study introduces a novel soft exosuit system integrating a controllable anchor mechanism and a single-motor, dual-drive transmission system. This integrated approach effectively mitigates the actuator redundancy inherent in multijoint assistance paradigms. The implemented system facilitates the sequential delivery of assistance for ipsilateral hip flexion and ankle plantar flexion within a single gait cycle through precisely timed, gait-phase-based control strategies. The article details the exosuit architecture, encompassing the controllable anchor, an integrated dual-drive actuation unit, an automatic tensioning system, and a biomechanically informed wearable textile interface. Addressing the significant challenge of modeling the complex, nonlinear dynamics governing soft exosuit-human interaction, a model-free adaptive control (MFAC) strategy employing a single-input-single-output (SISO) controller architecture was developed and implemented for precise assistance magnitude regulation. Experimental validation demonstrated significant biomechanical efficacy: surface electromyography (sEMG) signal amplitudes exhibited reductions ranging from 7.7% to 17.7% across key lower limb musculature. Furthermore, a 14.7% reduction in net metabolic rate was observed during loaded walking. These findings substantiate that the proposed system enhances biomechanical assistance performance while concurrently reducing metabolic energy expenditure.

Index Terms—Cable-driven mechanism, metabolic rate, model-free adaptive control (MFAC), soft exosuit.

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I. INTRODUCTION

WEARABLE exoskeletons have great potential in enhancing human movement functions and improving walking efficiency by providing external power according to user needs [1]. For healthy users, wearable soft exoskeletons are highly anticipated and effective in enhancing movement capabilities [2]. They can transfer auxiliary power to target joints through Bowden cables without interfering with the normal movement of the human body, enabling users to consume less energy during movement [3]. Most of the rigid exoskeletons show full application potential in terms of terrain adaptability [4], flexible actuators [5], and human-machine applicability [6], but soft exosuits demonstrate superior joint adaptability compared to rigid exoskeletons. One of the main goals of soft exosuit is to reduce the energy consumption of users during walking. According to the periodic changes in human gait, choosing to increase the number of driven joints or the different movements of a single driven joint is an effective strategy for achieving multijoint driven soft exosuit.

Soft exosuits utilizing Bowden cables for power transmission inherently operate as underactuated systems, typically requiring at least one dedicated actuator per assisted joint to provide tensile force [7], [8], [10]. While this approach has proven effective for assisting single joints such as hip and ankle [7], [8], scaling to multijoint assistance presents significant challenges in actuator count and system complexity. Strategies such as utilizing bidirectional rotation of a single actuator to assist flexion/extension of a single hip or extension of bilateral hips [9], or sequentially actuating knee extension and ankle plantarflexion within a gait cycle using one actuator per limb [11], demonstrate promising efforts to mitigate actuator redundancy and reduce system mass.

Recent research has begun integrating novel actuation and control strategies. Studies utilizing twisted string actuators (TSAs) have explored an adaptive control framework that employs an enhanced twin delayed deep deterministic policy gradient (TD3) reinforcement learning algorithm. This framework dynamically optimizes impedance parameters based on electromyography (EMG) and joint stiffness, enabling switching between robot-dominant, normal assistance, and human-dominant modes; it is also combined with a nonlinear disturbance observer for enhanced robustness [12]. Similarly, a model-free reinforcement learning-based mirror-adaptive impedance control (MAIC) framework was introduced for impedance optimization. This framework can switch between assistive/resistive modes in real time based on synchronized EMG and joint stiffness feedback [13].

To reduce actuator count, innovative approaches have been explored. Kim et al. demonstrated that a single actuator per limb

can sequentially drive knee extension and ankle plantarflexion assistance in sync with the gait cycle, effectively minimizing redundant actuators and system mass. While effective for specific, predefined multijoint sequences, this approach remains constrained by the fundamental limitation of Bowden cable systems: their fixed routing inherently prevents dynamic reconfiguration of assisted joints during operation [10], [14].

However, these approaches remain fundamentally constrained by the inherent limitations of Bowden cable systems: they can only transmit tensile force and, critically, their fixed routing paths rigidly couple actuators to specific anatomical anchor points [10], [14]. This fixed routing inherently limits the system's adaptability, preventing dynamic reconfiguration of the assisted joint during operation. Consequently, increasing the number of target joints inevitably necessitates a proportional increase in actuators [14], demanding innovative solutions to enhance flexibility without compromising functionality.

While strategies such as sequential joint actuation [11] reduce actuator count, they remain constrained by Bowden cables' fixed routing—fundamentally limiting dynamic joint reconfiguration during gait.

However, the position of the Bowden cable anchor is fixed, which means that the assisted joint cannot be changed. When the number of driven joints increases, the number of actuators will also increase accordingly. Therefore, innovative methods are needed to reduce the number of motors while maintaining functionality [14].

In response to the above problems, this article introduced a new type of multijoint assisted soft exosuit with controlled anchor points. By tightening and releasing the Bowden cable with a servo motor, the force transmission path of the Bowden cable is changed, thereby transferring the auxiliary power to the hip and ankle joints. In the human gait cycle, the flexion of the hip joint and the plantar flexion of the ankle joint show a clear sequence, with no overlap within a single cycle, and the peak joint torques at the hip and ankle occur at different phases of the gait cycle, enabling phase-separated assistance. At the same time, we have designed a new actuator, which uses two one-way bearings assembled in reverse on a single motor. When assisting one side of the hip or ankle joint, the Bowden cable on the opposite side remains stationary during unilateral joint actuation.

This new type of soft exosuit uses a single motor to assist four joints of the lower limb. The design maintains the existing force transmission path and introduces a controlled anchor, making it easy to switch between different joints under control. The exosuit is driven by Bowden cables, with four controlled anchors set above the knee joint and at the heel, and an automatic tensioner on the lateral side of foot. When driving the hip joint, the controllable anchor point above the knee joint will clamp the Bowden cable to ensure rigid transmission from the actuator to the anchor.

The main contributions of this article are as follows.

- 1) Based on the concept of altering the force transmission path through a controlled anchor, a novel soft exosuit featuring a controlled anchor, a new anchor positioning method, an automatic tensioning mechanism, and a drive unit integrated with one-way bearings. This innovative

soft exosuit can assist the hip and ankle joints within a gait cycle by clamping the Bowden cables at the anchor points while maintaining the original load transmission path.

- 2) A novel drive structure was introduced. By mounting two one-way bearings in reverse on a flange, when the exosuit assists one joint, the Bowden cable on the other side will not be released as the motor rotates. This novel driver structure solves the problem of nondrive control response delay in the control process.
- 3) For this soft exosuit system, an MFAC framework was implemented to address the inherent nonlinearities arising from soft tissue compliance and dynamic human-machine interactions, which typically degrade assistance trajectory tracking accuracy. The proposed control architecture enables precise joint torque delivery through real-time adaptation to time-varying biomechanical impedance, effectively compensating for nonlinear disturbances induced by anchor point displacement and biological tissue deformation during gait cycles.

The rest of this article is organized as follows. Section II introduces the design of the soft exosuit and discusses anchor-driven mechanisms for the soft exosuit. Section III presents auxiliary force profile and controller designs. Human experiments are conducted in Section IV. Finally, Section V concludes this article.

II. SYSTEM DESIGN

A. System Overview

The soft exosuit features a designed transmission path, using Bowden cable as the driving medium to accurately deliver assistive torques to the ankle and hip joints following the sequence of human gait cycles, shown in Fig. 1. The wearable components of this lower limb soft exosuit consist of a waist belt, two controlled anchors for controlling the hip joint, Bowden cable sheath, integrated inertial sensor mounted on the outer side of the shank, and a force output interface at the ankle joint. It also includes an integrated foot device with two ankle joint controlled anchors with an automatic tensioner.

The Bowden cable is led out from the waist sheath, runs down along the front side of the thigh, through the sheath on the side of the shank, and finally connects to the controlled anchor point at the ankle joint while being linked with an automatic coiler. In this multijoint driven soft exosuit system, we innovatively achieve effective assistance for plantar flexion at one side ankle joint and hip flexion using only one Bowden cable without changing the load path.

When assisting on one side's hip joint, clamping Bowden cable by controlled anchors allows to transfer of assistive force; when assisting the ankle joint, clamping the Bowden cable by controlled anchors allows for the transfer of assistive force. The triggering moment for clamping by controlled anchor is achieved through sensing human gait via an inertial measurement unit (IMU).

To mitigate the risk of Bowden cable rupture and prevent failure of the active anchor points during cable clamping, the motor output torque is constrained within the control system. Furthermore, the exosuit incorporates an automatic shutdown protocol triggered by



Fig. 1. Overview of soft exosuit: 1) waist belt; 2) Bowden cable; 3) load cell; 4) hip controlled anchor; 5) Bowden cable sheath; 6) actuator unit; 7) Bowden cable sheath; 8) IMU; 9) shank structure; 10) ankle controlled anchor; and 11) automatic tensioner.

abrupt changes in the force measured by the load cell. This specifically includes the rapid force decay to zero resulting from cable rupture. An emergency stop switch is also implemented to enable immediate full-system power cutoff, providing a manual safety override for the user in case of discomfort during operation. The battery is positioned at the user's waist. This waist-mounted location was selected to minimize interference with human motion [3] and facilitate the overall integration of the exosuit system.

B. Anchor Locations

Bowden cable, sutured fabric, and fixed anchor points to the human lower limb joints [15]. Due to the limitations of fixed anchor points, a single motor's forward and reverse rotation cannot directly drive multiple joints in the human body. Therefore, we have developed a controlled anchor system to replace the traditional fixed anchor, which assists in transmitting forces from the Bowden cable to the hip and ankle joints of the human body by clamping the Bowden cable.

When selecting controlled anchor locations, the anchors need to have appropriate rigidity to transmit forces. They should also maintain comfort to the greatest extent possible and reduce the pressure caused by fabrics [16]. From an anatomical perspective, human tissues are divided into three layers from superficial to deep: skin, fat, and muscle. The amount of fat directly affects tissue rigidity, with increased fat content enhancing interference with force transmission due to sliding between bone and skin. In the lower limb structure, there is less subcutaneous fat above the knee joint at the distal end of the thigh area with necessary rigidity suitable for the controlled anchor of the soft exosuit. Considering

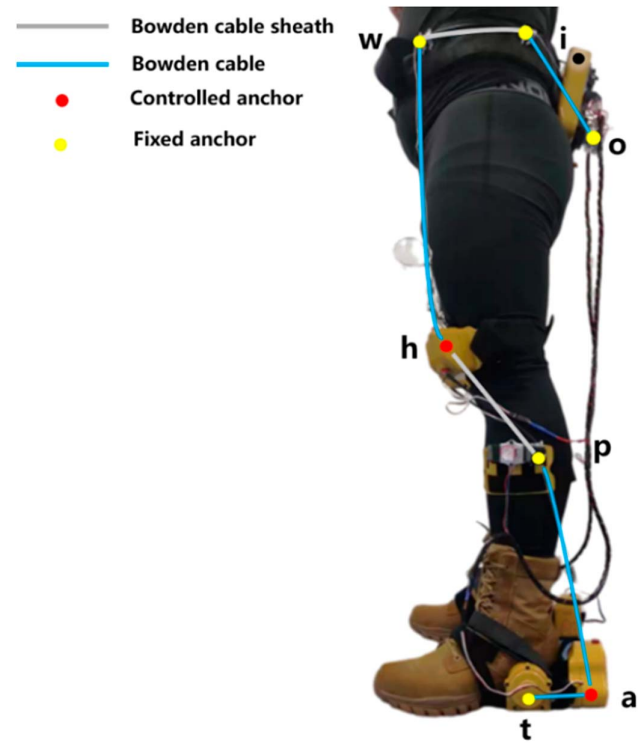


Fig. 2. Load path of soft exosuit.

different joint power ratios during walking processes, plantar flexion of the ankle plays a vital role during locomotion. Therefore, the heel positions, with low-fat content and high rigidity, can also serve as ideal anchor locations.

C. Load Path

To assist hip flexion and foot-ankle plantar flexion processes in sagittal plane movement on wearer's bodies, a carefully designed Bowden cable path shown was proposed, as shown in Fig. 2. Observing along the coronal axis, the Bowden cable is drawn from O point to the Bowden cable sheath on the belt, from i point to Bowden cable sheath, from w point to h point to Bowden cable, point h is the controlled anchor of the hip, point h to point p is the Bowden cable sheath, point a leads the Bowden cable from point p to the controlled anchor point of the ankle at point a, and finally enters point t, the position is the automatic tensioner.

The design intent behind this pathway encompasses two key aspects. First, it ensures that the assistive force in the Bowden cable effectively supports joint movements in the wearer's sagittal plane. This is achieved through specifically designed multiple entry-exit points that maintain parallel alignment during assisted muscular movements. Second, during level-ground walking, both ankle plantar flexion and hip flexion occur without temporal overlap, thereby contributing significantly to the respective power outputs of various leg joints. Meanwhile, the soft exosuit system should effectively support hip extension and foot-ankle plantar flexion, further facilitating smooth assistance through coordinated efforts. The exit points, located on the anterior side of the waist and the posterior side of the shank, along with controlled anchoring spots, ensure proper force transmission.

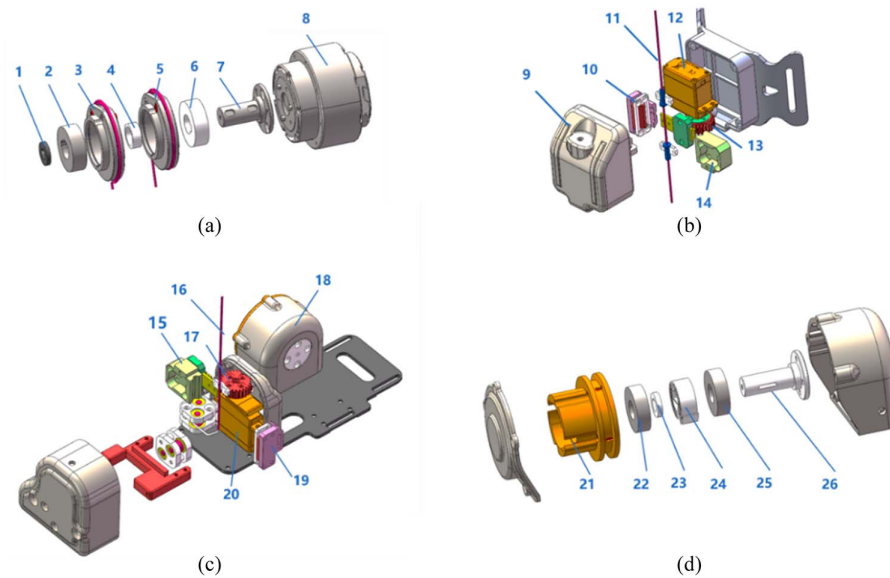


Fig. 3. Design of the unit of soft exosuit. (a) Actuator unit. (b) Hip controlled anchor. (c) Ankle controlled anchor. (d) Automatic tensioner. 1) Gasket of the actuator; 2) single row radial bearing I of the actuator; 3) pulley I; 4) axle sleeve of the actuator; 5) pulley II; 6) single row radial bearing II of the actuator; 7) flange of the actuator; 8) motor; 9) shell of hip controlled anchor; 10) rack clamping of hip controlled anchor; 11) Bowden cable; 12) servos of hip controlled anchor; 13) gear of hip controlled anchor; 14) baffle of the hip-controlled anchor; 15) baffle of the ankle controlled anchor; 16) Bowden cable; 17) gear of the ankle controlled anchor; 18) automatic tensioner; 19) rack clamping of the ankle controlled anchor; 20) servos of the ankle controlled anchor; 21) winding hinge; 22) bearing of the automatic tensioner; 23) axle sleeve of the automatic tensioner; 24) constant force spring; 25) bearing of the automatic tensioner; and 26) flange of the automatic tensioner.

D. Actuator Unit

In the design of a single motor-driven soft exosuit, a traditional solution is implemented to configure double cable pulleys on the same flange plate. This configuration allows the Bowden cable of the human joint on one side to slip easily when the motor begins driving, leading to excess cable overflow as the flange plate rotates, which creates a dead zone in the control process and results in control delays. To address this issue, an innovative bidirectional drive design was introduced, as illustrated in Fig. 3(a) and the size in Fig. 4(a). This design features two unidirectional bearings (CSK15PP, KIF, Japan) installed in opposite directions on the same flange plate, ensuring that when one side of the leg is driven by a motor, it operates according to the direction set by the unidirectional bearings, thereby preventing interference with the Bowden cables on the other side of the legs. This drive structure consists of a motor flange plate, two single-row radial bearings, an axle sleeve, two pulleys, and a gasket.

During unilateral assistance, the Bowden cable on the actuated side is tensioned by the motor driving the one-way bearing in its locked direction. Concurrently, the cable spool on the contralateral side remains stationary. To maintain baseline cable tension during nonassist phases and prevent interference with natural human movement, the system incorporates an automatic cable management mechanism.

E. Controlled Anchor

The controlled anchors at the hip and ankle joints are actuated by servo motors (DS3515, JUSHIDE, China) as illustrated in Fig. 3(b) and (c), the size of which is shown in Fig. 4(b) and (c),

where precision gears mounted on the motor output shafts engage with linear racks. This mechanism drives sliding blocks to rapidly secure Bowden cables through instantaneous clamping actions. Specifically, the selected servo motor, operating at 6 V, achieves a rotational speed of 0.1 s per 60°. With an initial gap of 1 mm between the two clamping blocks, a rotation of approximately 15° is sufficient to securely clamp the Bowden cable. This clamping action requires approximately 0.025 s.

Friction ridges are integrated onto each contact surface interfacing with the Bowden cable to enhance clamping friction. The cable remains clamped throughout the assistance phase. Additionally, guide holes are incorporated into the housing to facilitate smooth entry of the Bowden cable into the controlled anchor.

Specifically, the hip joint anchor clamps the Bowden cable during hip flexion assistance, while the ankle joint anchor performs similar clamping to facilitate plantar flexion assistance. This actuation strategy enables the soft exosuit to provide targeted joint assistance through selective cable constraint.

F. Automatic Tensioner

An automated tensioning device, featuring a winding hinge and a constant-force spring as depicted in Fig. 3(d) and the size in Fig. 4(d), maintains optimal pretension in the Bowden cables throughout the operation of the soft exosuit. This design not only guarantees consistent force transmission but also autonomously adjusts the cable length during the swing phase of gait cycles. The adaptive length compensation mechanism effectively reduces mechanical interference with natural human locomotion, thereby preserving physiological gait patterns while providing assistive torque.

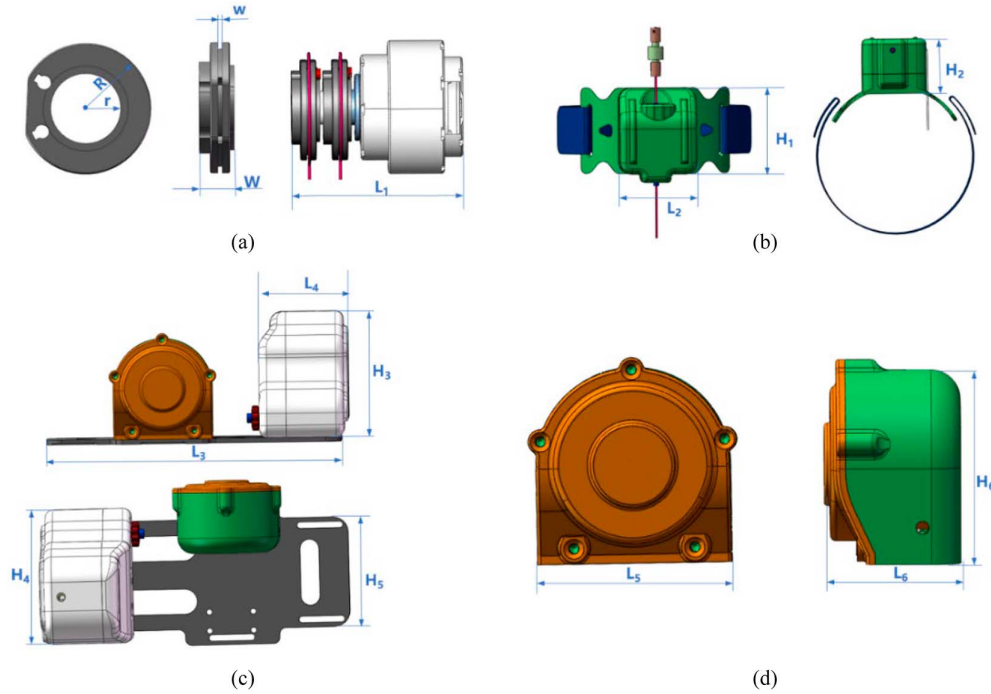


Fig. 4. Design size of the unit of soft exosuit. (a) Actuator unit. (b) Hip controlled anchor. (c) Ankle controlled anchor. (d) Automatic tensioner. $R = 30$ mm, $r = 16$ mm, $L_1 = 97$ mm, $W = 16$ mm, $w = 2$ mm, $L_2 = 65$ mm, $H_1 = 80$ mm, $H_2 = 55.5$ mm, $L_3 = 190$ mm, $L_4 = 57.5$ mm, $H_3 = 80$ mm, $H_4 = 85$ mm, $H_5 = 70$ mm, $L_5 = 64$ mm, $L_6 = 80$ mm, $H_6 = 65$ mm.

G. Sensors and Control System

Two different types of sensors are used for signal feedback and gait detection: The IMU (LPMS-2, Alube, China) is mounted on the left and right legs for detecting angular velocity and angular acceleration of the lower legs. The gait phase was detected by threshold method according to the angular velocity and angle of the shank sensor in the sagittal plane. The force sensor (AT8201, Euroda, China) is at the end of Bowden cable and measures the assist force. The soft exosuit is controlled by an STM32F407 microcontroller, which performs advanced calculations at a frequency of 1 kHz and communicates with the servo motor driver and sensor. The movement information of all three axes on IMU is transmitted through the controller area network (CAN) bus protocol. The CAN bus communication protocol is used to communicate between the 8-bit microcontroller and the servo motor driver. The force sensor transmits the force sensor signal to the main control board through the universal asynchronous receiver-transmitter (UART) interface through the force acquisition module. The controlled anchor is actuated via a pulse width modulation (PWM) generated by the microcontroller to clamp or release the Bowden cable within the controlled anchor. A 24 V battery with a capacity of 7 Ah powers the main control board and the force acquisition module.

III. CONTROLLER DESIGN

A. Assistance Force Profile Design

To provide continuous assistance to the hip and ankle joints of the human body through a soft exosuit within single gait cycle,

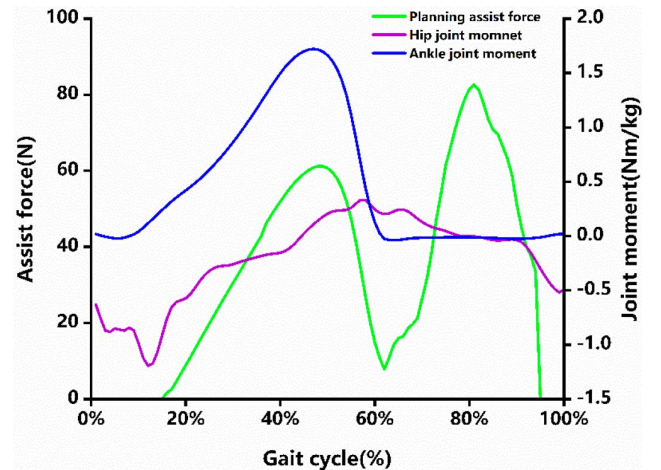


Fig. 5. Assistance profile and joint moment, 0% is the heel contact.

we used a public dataset [17] and analyzed the biomechanical torque profiles of the human hip joint and ankle joint within one gait cycle, as shown in Fig. 5. The plantarflexion torque of the ankle joint is mainly generated between 11% and 64% of the gait cycle, with peak torque occurring at approximately 49%. The biomechanical flexion torque of the hip joint appears between 33% and 77% of the gait cycle, reaching its peak at around 52%. The plantar flexion of the ankle joint and flexion of the hip joint do not overlap in time during a complete gait cycle, allowing for designing assistance torque profiles based on biomechanical torque profiles and sequence design by offsetting the flexion biomechanical torque profile of the hip joint to assist with hip flexion process.

We predefined an assistance torque profile based on the required joint torques for ankle plantar flexion and hip flexion.

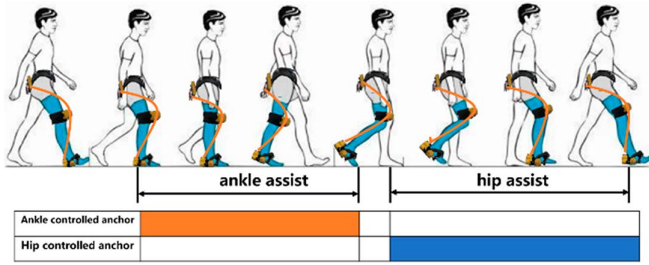


Fig. 6. Assistive sequence diagram of the soft exosuit. The orange bar represents the ankle controlled anchor work, and the blue bar represents the hip controlled anchor work.

Calculating this assistance profile involved cubic data fitting, determining thigh length as well as the distance from the ankle joint to the pivot point, resulting in finalizing an assistance profile. This profile consists primarily of ankle plantar flexion from 15% to 61% GC (gait cycle) and hip flexion from 61% to 90% GC. Assistance torques are nearly zero during periods from 0%–15% GC and from 90%–100% GC. The various parts of the auxiliary assist moment profile are smoothly connected.

We explained the order of assistance in Fig. 6. The figure illustrates the actuation sequence for the assisted joints on one side of the body. During this process, the unassisted contralateral side performs normal walking. Following the completion of assistance to the first side, actuation proceeds to the hip and ankle joints of the opposite side. The assistance phases for both sides occur in a sequential manner without temporal overlap or conflict. Crucially, one side remains unassisted while the other is being actuated. This sequential bilateral actuation strategy thereby enables the actuation of all four lower limb joints using a single actuator.

The actual joint torque approaches zero at approximately 80% of the moment of hip. This coincides with the swing phase of the human gait cycle, where lower limb motion becomes primarily governed by the inertial properties of the leg during forward progression. This natural swinging motion corresponds to the hip joint flexion phase. Due to inertial motion during this swing, the hip joint torque is inferred to be approximately zero. Concurrently, the ankle joint is in the tip-off phase (no longer in ground contact), and its torque is similarly inferred to be approximately zero.

In contrast, the peak hip joint torque occurs at heel-off initiation. This maximum torque arises from synergistic joint interactions, while the swing phase torque is predominantly inertial.

By using the force profile along with sequential driving mode, the motor provides assisting torques sequentially to each leg's ankles and hips, improving system efficiency.

B. Controller

In the previous design of the soft exosuit controller, an iterative learning approach was used in conjunction with a human-machine coupled stiffness model to adapt to variations in leg length and gait of different individuals. However, obtaining the human-machine stiffness model in an actual soft exosuit system is based on fitting

experimental data. Sometimes, the fitted model may not fully adapt to individuals. Additionally, nonlinear elements are present in soft exosuit systems due to human motion, skin sliding, and fabric elasticity deformations that cannot be completely compensated for by stiffness models.

The soft exosuit only uses soft fabrics to connect to the human body. Due to the differences in human limbs and the stiffness changes of the fabric itself, there are a large number of nonlinear links in the process of tight coupling between the soft exosuit and the human body. And because the soft exosuit does not have rigid structural parts, and the soft exosuit we designed only has inertial sensors installed on the shanks, the Euler-Lagrange method commonly used in exoskeletons is not suitable for modeling our soft exosuit.

Since we installed a force sensor on the end of Bowden cable to measure the force in the Bowden cable, using a single leg as the observation object can be considered as a SISO system, and the input and output can be measured, solving problems that cannot be accurately modeled.

Therefore, we use a compact form dynamic linearization model free adaptive control method to control soft exosuit using only input and output data from the system. This approach can overcome uncertainties and disturbances with strong robustness toward disturbances and unknown model dynamics [18]. Since our soft exosuit uses force sensors to control input and target objectives, it can be considered as a nonlinear single-input single-output control system that satisfies three assumptions of SISO-MFAC:

Assumption 1: The partial derivatives of $F(k+1)$ with respect to control inputs $F_a(k)$ and outputs $F(k)$ are continuous. $F(k)$ represents the output force of the system at the next moment, and $F_a(k)$ is the input of the system at time k .

Assumption 2: The soft exosuit system is a generalized Lipschitz system. That is, for any moment $k_1 \neq k_2$, with $k_1, k_2 \geq 0$, $F_a(k_1) \neq F_a(k_2)$, $|F(k_1+1) - F(k_2+1)| \leq b|F_a(k_1) - F_a(k_2)|$, $b > 0$ is a positive constant.

Assumption 2 means that if the energy change of the control input is at a finite level, the energy change inside the actual system cannot reach infinity. In practical applications, many practical motion control systems and physical processing satisfy such properties.

Assumption 3: For any given bounded desired output signal $F_d(k+1)$, there always exists a bounded control input signal $F(k+1)$ such that, when driven by this input signal, the system's output equals $F_d(k+1)$.

Assuming that the input and output of the soft exosuit system at moment k are $F_a(k)$, $F(k)$, represented as follows:

$$F(k+1) = f(F(k), F(k-1), \dots, F_a(k), F_a(k-1)). \quad (1)$$

The soft exosuit system satisfies Assumptions 1–3, then the soft exosuit system (1) can be approximately equivalent according to the following CFDL data model:

$$\Delta F(k+1) = \phi(k) \Delta F_a(k) \quad (2)$$

where $\Delta F(k+1) = F(k+1) - F(k)$, and $\Delta F_a(k+1) = F_a(k+1) - F_a(k)$.

Consider the following PPD (pseudopartial derivative) estimation criteria function:

$$J(\phi(k)) = |F(k) - F(k-1) - \phi(k)\Delta F_a(k-1)|^2 + \mu|\phi(k) - \hat{\phi}(k-1)|^2. \quad (3)$$

$\mu > 0$ denotes the weight factor. The parameter μ is crucial in MFAC for ensuring numerical stability and robustness. Its primary role is to regularize the pseudo partial derivative update process, preventing division-by-zero errors when input changes are small. Furthermore, μ balances the sensitivity and stability of parameter updates and, together with η , jointly dictates the characteristics of the learning process; in this article, $\mu = 0.001$.

For (3), we get the estimation value of PPD by finding the extreme value

$$\hat{\phi}(k) = \hat{\phi}(k-1) + \frac{\eta\Delta F_a(k-1)}{\mu + \Delta F_a(k-1)^2}(\Delta F(k) - \hat{\phi}(k-1)\Delta F_a(k-1)). \quad (4)$$

$\eta \in (0,1)$ is the step size factor added to enhance the flexibility and generality of the algorithm.

Consider the following control input criteria function:

$$J(F_a(k)) = |F_d(k+1) - F(k+1)|^2 + \lambda|F_a(k) - F_a(k-1)|^2. \quad (5)$$

$\lambda > 0$ denotes the weight factor, the first term is introduced to make the output of the system consistent with the desired output, and the second term is the regular term, which penalizes big input change to reduce the input energy and oscillations.

Substituting (2) into criterion function (5), taking the derivative of $F(k)$ and making it equal to zero, the following control algorithm can be obtained, we get the output $F(k)$

$$F(k) = F(k-1) + \frac{\rho f(k)}{\lambda + |\phi(k)|^2} (F_d(k+1) - F_a(k)) \quad (6)$$

where $F(k)$ represents the output force of the system at the next moment, $F_a(k)$ is the input of the system at time k , $\phi(k)$ represents the PPD estimation value of the system, and $\rho \in (0,1)$ is the step size factor, which is added to make the control algorithm more general.

According to the interaction characteristics between the soft exosuit and the human body, the admittance controller was used to express the take-up amount of Bowden cable (the rotation angle of the motor) and the auxiliary force output of the soft exosuit and the second-order differential equation

$$F = M_d(\ddot{x}_a - \ddot{x}_d) + B_d(\dot{x}_a - \dot{x}_d) + K_d(x_a - x_d) \quad (7)$$

where F represents the auxiliary force output of the soft exosuit; M_d , B_d , and K_d represent the desired target inertia, target damping, and target stiffness; $\ddot{x}_a$, $\dot{x}_a$, and x_a , represent the actual motor acceleration, the actual motor speed, and the actual motor rotation angle; and $\ddot{x}_d$, $\dot{x}_d$, and x_d indicate the expected motor

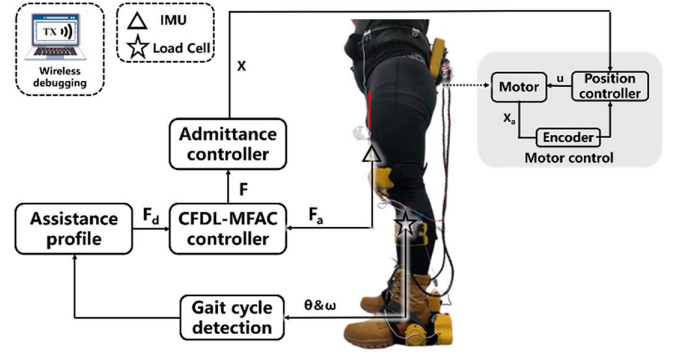


Fig. 7. Implementation of control system.

rotation acceleration, the desired motor rotation speed, and the desired motor rotation angle.

In the actual tight coupling system between the soft exosuit and the human lower limb, the relationship between the auxiliary force output of the soft exosuit, and the motor rotation angle is expressed as

$$F = G(x_d - x_a) \quad (8)$$

where G represents the admittance model and F represents the auxiliary force output of the soft exosuit.

The position information of the motor can be perceived in real-time through the encoder on the motor, so the expression of the admittance controller can be obtained through (7) and (8)

$$F = M_d\Delta\ddot{x} + B_d\Delta\dot{x} + \Delta K_d\Delta x. \quad (9)$$

Equation (9) yields an admittance model of the motor rotation angle and the force exerted on the target human joint

$$\Delta X = \frac{F}{M_d\Delta\ddot{x} + B_d\Delta\dot{x} + K_d}. \quad (10)$$

Then, the take-up amount of the motor is calculated by (10), and the take-up amount ΔX of the motor is then input into the inner loop of the motor position control controlled by a PID controller, and finally, the tracking of the expected force profile is realized.

The schematic diagram of the multijoint soft exosuit control system used is shown in Fig. 7. The control system consists of a gait recognition system, an auxiliary force profile generator, SISO-MFAC, and an admittance controller. By detecting gait events using IMU angle and angular velocity signals to load auxiliary force profiles, the SISO-MFAC controller can realize the actual motion trajectory of the exosuit and accurately track the expected motion trajectory. The admittance controller converts force signals into motor position signals and achieves accurate position tracking inside the motor through the motor's encoder. Using a soft exosuit with controlled anchor points as the controlled object aids with hip and ankle joints for healthy individuals.

IV. EXPERIMENT AND RESULTS

A. Experiment Protocol

Four healthy male subjects (age: 25 ± 1.5 years; body mass: 90 ± 5.6 kg; height: 1.74 ± 0.03 m) participated in a physiological

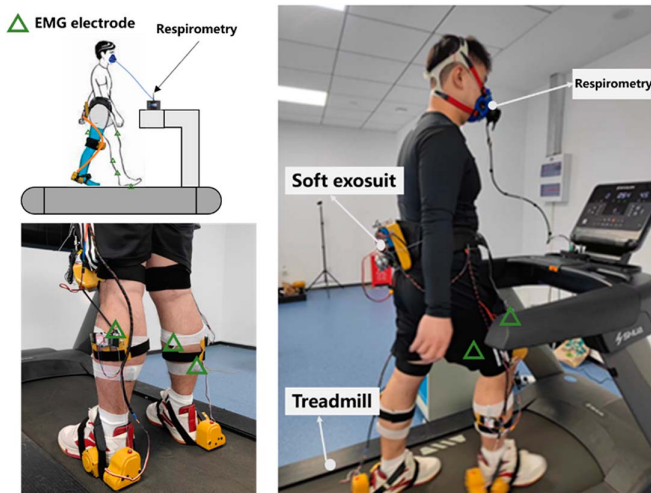


Fig. 8. Experimental setup. Testers were asked to walk on a treadmill at a speed of 4.5 km/h. EMG electrodes and K5 were securely affixed to measure the physiological changes.

evaluation experiment of a soft exosuit to validate its efficacy. The study protocol involving human subjects was approved by the Ethics Committee of Northeastern University, and written informed consent was obtained from all participants before the experiment.

As illustrated in Fig. 8, the experimental setup consisted of a treadmill walking task at 1.25 m/s. Surface electromyography (sEMG) electrodes and a portable spirometer were securely attached to participants using medical adhesive and elastic bandages. During the preexperimental session conducted 24 h before formal testing, participants received detailed instructions on soft exosuit operation and completed a 30-minute familiarization trial to ensure proper device utilization.

The main experiment employed a three-condition crossover design: 1) powered soft exosuit (exo-on); 2) unpowered soft exosuit (exo-off); and 3) no exosuit (without exo). Each subject completed one 5-minute trial per condition in randomized order, with 15-minute rest intervals between trials to mitigate muscle fatigue effects. All experimental sessions were conducted under standardized laboratory conditions (ambient temperature: $23 \pm 1^\circ\text{C}$; relative humidity: $50 \pm 5\%$).

B. Metabolic Rate Experiment and Results

The volumetric exchange of oxygen and carbon dioxide during respiration varies with metabolic demand, as carbon dioxide production serves as a direct byproduct of metabolic processes during physical activity. Consequently, the assessment of gas composition and volumetric parameters in exhaled breath provides a scientifically valid methodology for evaluating metabolic activity [19].

As an objective physiological indicator, the human metabolic rate effectively quantifies the biomechanical impact of soft exosuit interventions on human locomotion. In this study, metabolic measurements were conducted using the COSMED K5 portable cardiopulmonary testing system (COSMED Srl, Italy), which employs Bluetooth-enabled telemetry for real-time

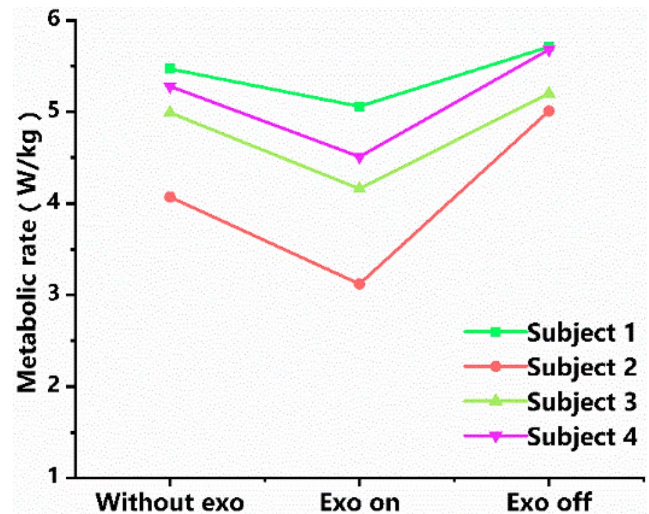


Fig. 9. Metabolic results of each subject.

data acquisition and subject monitoring. This instrument facilitates breath-by-breath analysis of the expired gas composition, enabling precise calculation of metabolic fluctuations during dynamic movement protocols.

The system's high-precision sensors allow accurate quantification of oxygen consumption (V_{O_2}) and carbon dioxide production (V_{CO_2}), parameters essential for computing metabolic energy expenditure through the Brockway Equation

$$E_{MET}(W) = (c_1 * V_{O_2} + c_2 * V_{CO_2})/60 \quad (11)$$

where c_1 and c_2 are 16.59 and 4.51, respectively, V_{O_2} is the volume of inhaled oxygen, and V_{CO_2} is the volume of exhaled carbon dioxide.

Fig. 9 illustrates the average metabolic rate for the four subjects across three conditions: without exo, exo on, and exo off. For measures that showed significant trends, we performed a Wilcoxon signed-rank test to examine differences between groups and identify the effect of SPURS (significance level $\alpha = 0.05$; SPSS Statistics, IBM). Compared to the exo off, the average metabolic rate decreased by 21.85% ($p = 0.008$) under the exo on condition. Compared to the without exo, the average metabolic rate decreased by 14.7% ($p = 0.024$) under the exo on condition, as shown in Fig. 10.

The metabolic experimental data demonstrate that this portable soft exosuit provides substantial biomechanical assistance to the wearer while effectively reducing energy expenditure during physical activities.

C. sEMG Experiment and Results

sEMG signals were recorded under two motor tasks: quiet standing and treadmill walking. For the standing protocol, participants maintained an upright posture on level ground for 1 min with arms relaxed at their sides, during which baseline sEMG signals were acquired.

The walking trials comprised three sequential conditions: 1) unimpeded walking (without exosuit); 2) exosuit-assisted walking with hip and ankle joint torque augmentation (exo on); and 3) exosuit-unpowered walking (exo off). Participants first

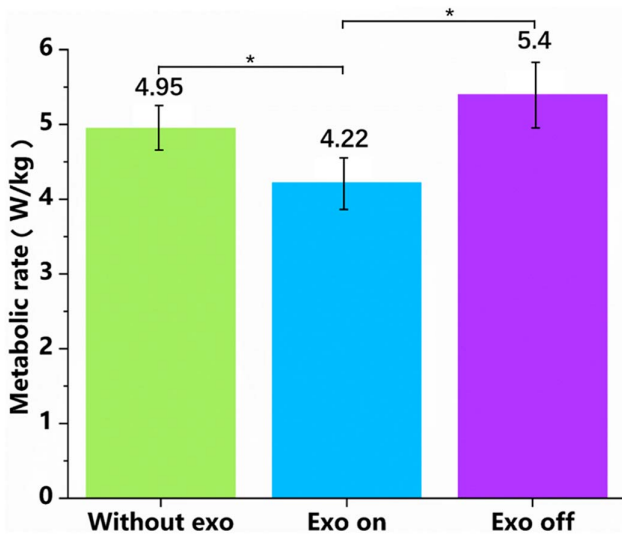


Fig. 10. Metabolic test results. Average metabolic results. All data were adjusted by subtracting the metabolic rate measured under the standing condition before calculating the metabolic change. The error bands represent SD.

completed a 5-minute treadmill acclimatization period at 1.25 m/s to establish steady-state gait kinematics before data collection. Continuous sEMG recordings were obtained during the final 2 min of each 5-minute walking bout to ensure kinematic stability. To control for circadian variations and muscle recovery effects, all experimental conditions were completed within a single session.

During walking, the lateral gastrocnemius (LG), medial gastrocnemius (MG), and soleus (SOL) drive the plantar flexion of the ankle joint, hip flexion is mainly driven by the semitendinosus (SEM) and rectus femoris (RF) muscles. The location of the sEMG sensor on the human body is shown in Fig. 11. The LG is located on the lateral side of the largest muscle mass on the back of the leg; the MG is located on the medial side of the leg, and the entire gastrocnemius formed by both is the basis for walking and various postures. The gastrocnemius and SOL are mainly responsible for plantar flexion of the ankle and provide a large amount of propulsion in walking, running, and jumping. The SEM, a spindle-shaped muscle at the back and inside of the thigh, is responsible for stretching the buttocks and flexing the knees. The RF, which is part of the quadriceps femoris group, is a bidirectional muscle located in the upper and middle thighs and is primarily responsible for knee extension and hip flexion movements during exercise.

This study uses LG, MG, SOL, SEM, and RF as experimental muscles commonly used in normal walking as research objects [20]. Experimental muscles for muscle fatigue experiments. For the results of muscle fatigue experiments, we use root mean square (rms), the following equation as an evaluation index for muscle fatigue degree:

$$\text{EMG}_{\text{rms}}(n) = \sqrt{\sum_{i=n-N+1}^n \text{sEMG}(i)^2}. \quad (12)$$

To visually illustrate the degree of reduction in muscle fatigue after wearing a soft exosuit, a set of comparative histograms is shown in Fig. 12.

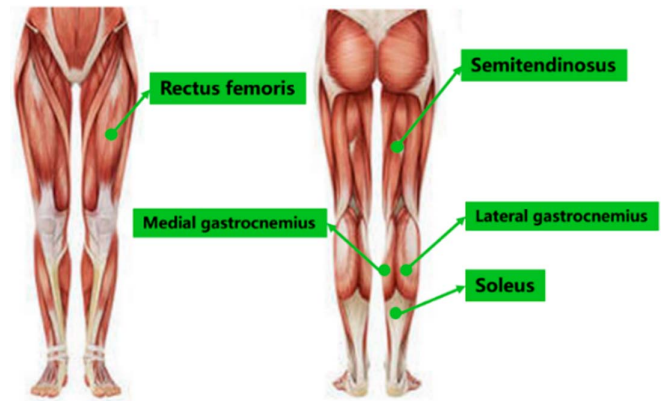


Fig. 11. Surface EMG sensor location.

The muscle activity of the LG, MG, SEM, SOL, and RF was characterized using sEMG signals measured with wireless electrodes (16, Cometa). The signals were initially filtered through a high-pass filter with a cutoff frequency of 20 Hz, rectified, and subsequently filtered through a low-pass filter with a cutoff frequency of 6 Hz. All results were averaged over the continuous stable strides during each trial and normalized against those obtained under normal walking conditions. For measures that showed significant trends, we performed a Wilcoxon signed-rank test to examine differences between groups and identify the effect of SPURS (significance level $\alpha = 0.05$; SPSS Statistics, IBM).

Electromyographic assessment revealed reductions in muscle activity across the targeted musculature during soft exosuit assisted trials, compared to the unassisted baseline. Specifically, the gastrocnemius lateralis, gastrocnemius medialis, SOL, RF, and SEM exhibited reductions of 7.7% ($p = 0.008$), 8.3% ($p = 0.008$), 17.1% ($p = 0.008$), 9.7% ($p = 0.008$), and 17.7% ($p = 0.008$). These findings demonstrate the efficacy of the soft exosuit in attenuating muscular exertion during a dynamic task.

D. Controller Experiment Results

The primary control objective for the soft lower limb exoskeleton is the precise temporal delivery of assistive forces according to a predefined profile throughout the gait cycle.

Graphical analysis was employed to evaluate the exosuit's assistive force tracking performance during wearer operation. As depicted in Fig. 11, significant temporal overlap is observed between the desired assistive force profile (blue dashed line) and the experimentally measured force (red solid line) within the two critical peak force regions associated with hip and ankle joint actuation. This close alignment demonstrates the exosuit's efficacy in delivering the targeted peak assistive forces as designed, thereby providing biomechanically appropriate support during gait phases demanding maximal joint torque.

Accurate dynamic modeling of the exoskeleton system is inherently challenging due to its nonrigid structure, compounded by nonlinearities arising from human soft-tissue deformation and friction within the Bowden cable transmission system. Model-free adaptive control (MFAC) effectively addresses this core challenge by eliminating the dependency on a priori mathematical models (whether derived from first principles or system

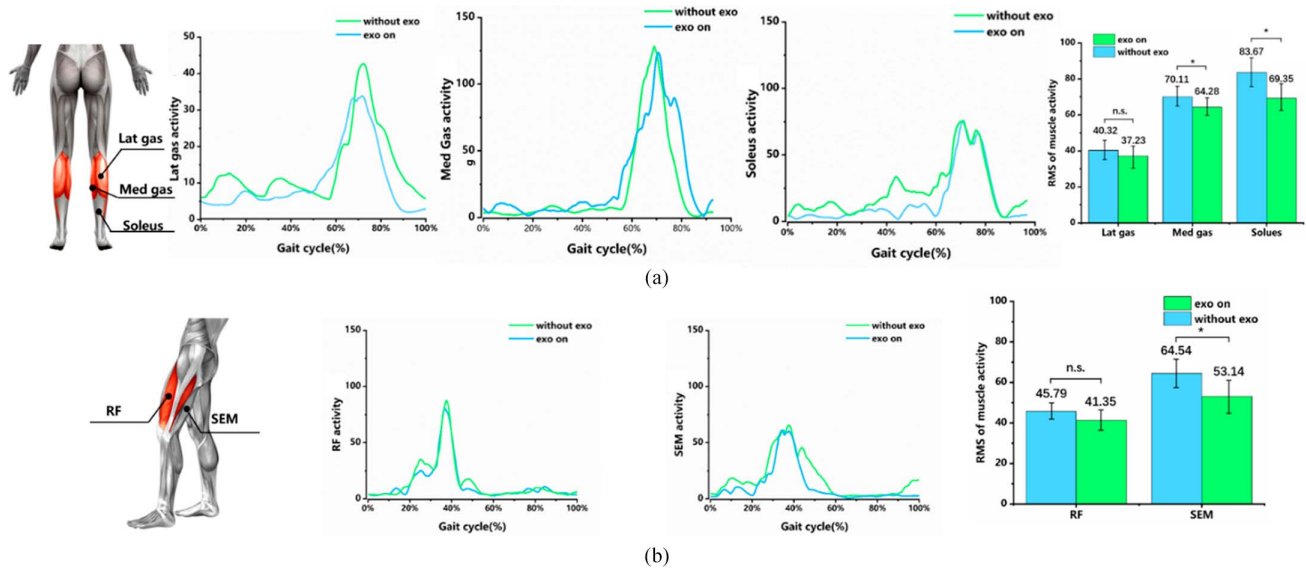


Fig. 12. Left: the primary muscles involved in the hip joint and ankle joint. Mid: the muscle activity for the lat gas, med gas, SOL, RF, and SEM over the gait cycle averaged across all participants for each experimental condition. All data were normalized by the muscle activity under the without exo condition. (a) Muscle activity results of lat gas, med gas, and SOL. (b) Muscle activity results of RF and SEM. The error bands represent SD.

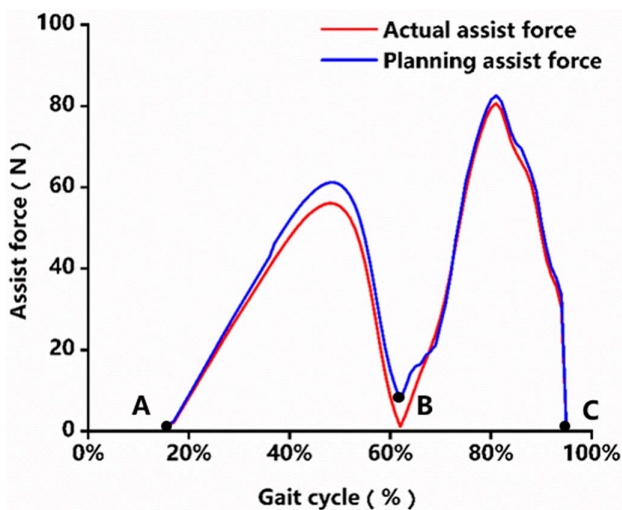


Fig. 13. Assistance profile result, point A is the moment when the auxiliary force begins to assist the ankle plantar flexion, point B is the moment when the auxiliary force ends the help on the ankle joint and begins to assist the hip joint, and point C is the moment when the hip joint assists the end, and the actual force and the designed assist force coincide in shape.

identification). MFAC operates solely using real-time input/output data to learn the system dynamics online and continuously adapt the control law. This approach renders MFAC intrinsically suitable for highly nonlinear systems where accurate modeling is impractical. In contrast, the performance of proportional-integral-derivative (PID) control is critically dependent on accurate system characterization. For strongly nonlinear systems such as the soft exosuit, identifying a single set of fixed PID parameters that ensure robust performance across all operating conditions proves exceptionally difficult.

Fig. 13 illustrates the force tracking under MFAC control. The target force profile (blue dashed line) and the measured force (red solid line) exhibit substantial temporal correspondence during the critical hip and ankle peak actuation phases,

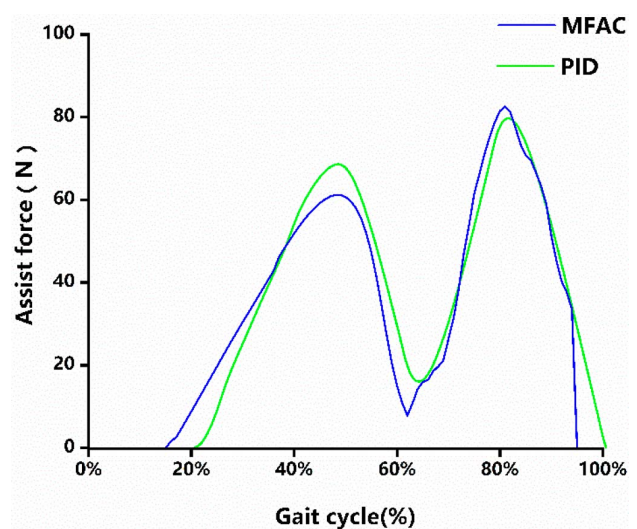


Fig. 14. PID contrast assistance profile result, the PID has a certain delay at the beginning and end of power assistance.

achieving a root mean square error (RMSE) of 2.82 N. This result further confirms the system's capability to deliver the intended peak biomechanical support when required.

Conversely, the PID-controlled system exhibits a consistent temporal delay equivalent to 5% of the gait cycle relative to the MFAC controller (visually apparent in Fig. 14). This temporal misalignment contributes to a degraded tracking performance, quantified by a higher RMSE of 3.51 N between the desired and measured force profiles.

V. DISCUSSION

This study introduces a novel soft exosuit system incorporating a controlled anchor mechanism and a single-motor, dual-drive transmission. The system demonstrated significant reductions in

TABLE I
COMPARISON OF EXISTING SOFT EXOSUIT

Study	Assist Target	Weight (kg)	Portability	Physiological Effectiveness
This work	Ankle PF and Hip Flex.	3.2	Portable	14.7% reduction in gross metabolic rate
Kim et al. [22]	Hip Flex	2.31	Tethered	9.6% reduction in gross metabolic rate
Orekhov et al. [23]	Ankle PF and DF	2.07	Portable	No significant change observed in gross metabolic rate
Quinlivan et al. [24]	Ankle PF and Hip Flex.	0.72 each leg	Tethered	3.59% reduction in gross metabolic rate
Miller et al. [25]	Ankle PF	1.10 each leg	Tethered	7.7% reduction in gross metabolic rate

metabolic cost (14.7%) and muscle activation (7.7%–17.7%). While these results validate the biomechanical efficacy of the proposed design and the MFAC strategy, several limitations and future research directions warrant consideration.

First, physiological evaluations were conducted on a cohort of four healthy male participants (age: 25 ± 1.5 years; BMI: 29.7 ± 1.8 kg/m²). Although statistically significant reductions in metabolic and sEMG metrics were observed, the cohort homogeneity limits the generalizability of the findings. Future studies should incorporate more diverse participant groups, including females, individuals across a wider age range, and varying BMIs. This is crucial for assessing the exosuit's adaptability to anthropometric variability and avoiding overgeneralization of performance claims. Table I presents a comparative analysis between the proposed exosuit and existing exoskeleton systems. While our design exhibits a marginally higher mass, physiological assessment results demonstrate that the device achieves reductions in metabolic cost and muscle activity comparable to those of prior designs.

Second, the total exosuit mass (3.2 kg) is consistent with contemporary designs. Power is supplied by a single 24 V, 7 Ah battery. The system's average electrical power consumption over a complete gait cycle is 10 W, resulting in an operational battery life exceeding 6 h during assisted walking. Additionally, the exosuit fabric's breathability and overall wear comfort were subjectively assessed via questionnaires [21] administered to the four testers. Results indicated favorable breathability; however, localized skin pressure was noted despite the use of anti-slip materials at limb interfaces. Future iterations will utilize more comfortable and breathable materials for limb attachment components. Quantitative comfort assessments employing standardized metrics should be integrated into future evaluations to optimize textile interface design and load distribution. The battery is mounted at the waist, positioned near the body's center of mass to minimize its inertial effect on locomotion and enhance system integration.

While the waist-mounted battery minimizes interference with leg dynamics, it may induce compensatory torso movements. Redistributing mass components—such as partitioning the battery pack onto the thigh/shin segments and utilizing lighter composite materials—could reduce lumbar inertial loading and improve comfort during prolonged operation.

The system implements a multilayer safety architecture to mitigate operational risks. Motor torque limits serve as the primary constraint, actively restricting actuator output to prevent forces that could cause Bowden cable rupture. Furthermore, an automatic shutdown mechanism is triggered in real time upon detecting rapid or abrupt force decay via an integrated load cell

providing continuous force feedback. For immediate intervention, a hardwired emergency cutoff switch, prominently positioned, enables instantaneous system deactivation independent of the control logic. These integrated features—torque limiting, automatic shutdown, and emergency cutoff—collectively establish a baseline operational safety framework.

Currently, assistance is provided solely in the sagittal plane (hip flexion, ankle plantarflexion). Extending functionality to address frontal/coronal plane demands, such as knee stabilization during stance and mediolateral balance control, would enhance utility on uneven terrain. This could involve integrating an assistive cable routing/adduction module to provide knee abduction-adduction moments. Enhanced gait phase detection is also proposed, leveraging the fusion of IMU data with machine learning techniques to improve terrain-adaptive gait segmentation using existing shank-mounted sensors. Finally, the MFAC framework could be expanded to dynamically modulate joint stiffness in response to real-time movement uncertainty.

VI. CONCLUSION

This study presents a novel multijoint lower limb soft exosuit system featuring an innovative controlled anchor integrated with Bowden cable clamping, enabling dynamic switching of assistive modes between hip and knee joints while maintaining consistent force transmission pathways. The prototype demonstrates breakthrough mechanical design through a single bidirectional motor driving four-joint coordinated motion, coupled with a biomechanically informed assistive force control model aligned with human joint torque characteristics. The control architecture implements a model-free adaptive algorithm that achieves high-precision tracking of assistive force profiles, circumventing the traditional reliance on intricate dynamic models. Physiological evaluations involving four participants under three conditions (exo-on, exo-off, and without exo) revealed reductions in sEMG and metabolic cost by 14.7% reduction, confirming the system's efficiency in enhancing human locomotor efficiency.

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